

student mode offers a free account, while the enterprise version is available for factories or industrial use.

After creating an Edge Impulse account, the Edge Impulse Data Forwarder must be installed on the PC. For macOS, Linux, or Windows, the installation can be completed using the terminal (cmd for Windows) by running the following commands sequentially:

```
- Installing Node.js:
```

```
node -v
npm -v
- Installing Edge Impulse CLI:
npm install -g edge-impulse-cli
- Setting up the Data Forwarder:
edge-impulse-data-forwarder
```

Connecting IndusBoard Coin to Edge Impulse

The IndusBoard Coin must be connected to Edge Impulse to collect sensor data. The data forwarder

code is required for this purpose. The code must be modified to extract data from the relevant sensor for other sensors like light, environment, air quality, or similar. In the loop, sensor data can be sent to the serial port. For multiple sensor values, each can be separated using:

```
Serial.print('\t');
```

For example, accelerometer data with three values (x, y, z axis) can be separated after each axis data using:

```
Serial.print('\t');
```

The data forwarder code should be prepared and modified according to the requirements. The code can be downloaded from *electronicsforu.com*. Detailed guidance on the data forwarder code is available at [Edge Impulse CLI - Data Forwarder] (<https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-data-forwarder>). Fig. 2 shows the Edge Impulse data forwarder code.

After uploading the data forwarder code and the IndusBoard to your laptop's USB port, open the terminal for Linux or macOS (Cmd for Windows), then run the

```
Last login: Mon Dec 9 21:17:31 on ttys000
ashwini@Ashwinis-MacBook-Air ~ % sudo edge-impulse-data-forwarder --clean
Password:
Sorry, try again.
Password:
Edge Impulse data forwarder v1.24.0
(node:3269) [DEP0040] DeprecationWarning: The 'punycode' module is deprecated. Please use a userland alternative instead.
(Use 'node --trace-deprecation ...' to show where the warning was created)
? What is your user name or e-mail address (edgeimpulse.com)? ashwini.sinha@efy.
in
? What is your password? [hidden]
Endpoints:
  Websocket: wss://remote-mgmt.edgeimpulse.com
  API:       https://studio.edgeimpulse.com
  Ingestion: https://ingestion.edgeimpulse.com

[SER] Connecting to /dev/tty.usbmodem01
[SER] Failed to connect to /dev/tty.usbmodem01 retrying in 5 seconds Error: Resource busy, cannot open /dev/tty.usbmodem01
[SER] You might need 'sudo' or set up the right udev rules
[SER] Failed to connect to /dev/tty.usbmodem01 retrying in 5 seconds Error: Resource busy, cannot open /dev/tty.usbmodem01
[SER] You might need 'sudo' or set up the right udev rules
[SER] Failed to connect to /dev/tty.usbmodem01 retrying in 5 seconds Error: Resource busy, cannot open /dev/tty.usbmodem01
[SER] You might need 'sudo' or set up the right udev rules
[SER] Serial is connected (0)
[WS ] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS ] Connected to wss://remote-mgmt.edgeimpulse.com

? To which project do you want to connect this device? Ashwini / gest
[SER] Detecting data frequency...
[SER] Detected data frequency: 50Hz
? 3 sensor axes detected (example values: [-3.81,3.56,8.68]). What do you want to call them? Separate the names with ',':
```

Fig. 3: Edge Impulse connecting with IndusBoard using the data forwarder in the terminal

DEVICE	ID	TYPE
 0 Connected to data acquisition (Sensor with 3 axes (x, y, z))	0	DATA_FORWARDER

Fig. 4: Device connection status in Edge Impulse

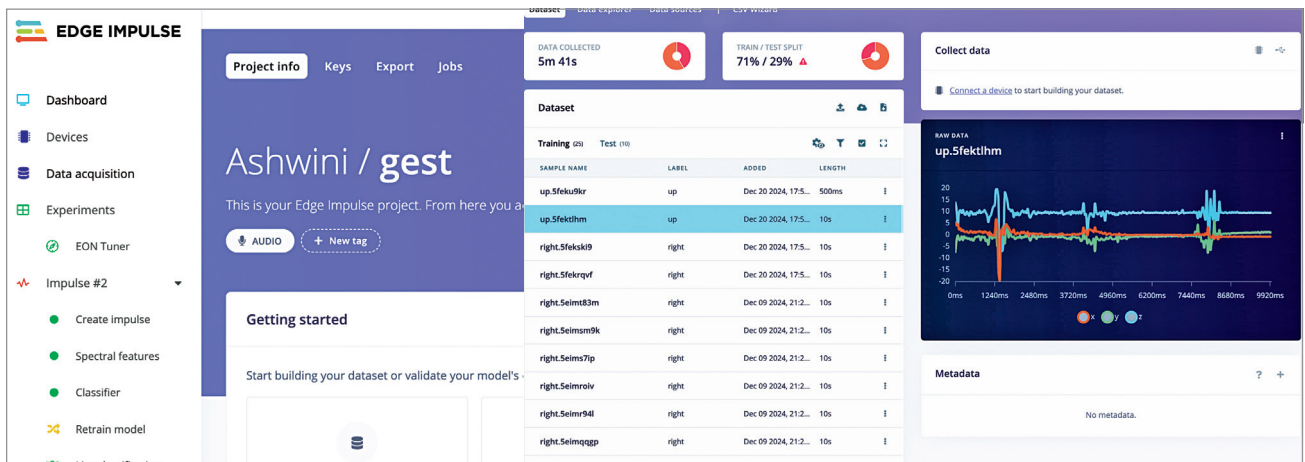


Fig. 5: Dataset collection